Code-Layout, Readability and Reusability

Date	28 September 2026
Team ID	SWTID-2026-3161
Project Name	Script-Controlled ACL – Restrict Record Access Based on Field Value
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	4-space indentation used in all ACL scripts and Script Includes.
2	Proper File Structure	Files and folders are logically organized	Yes	ACL scripts, Script Includes and test scripts kept in separate, clearly named records/folders.
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Names like userRoles, recordFieldValue and hasAccess describe their purpose.
4	Function / Method Names	Functions are descriptively named	Yes	Functions such as checkFieldAccess() and isUserAllowed() describe what they do.
5	Code Comments	Inline and block comments explain logic	Partial	Main ACL condition logic is commented; a few helper functions need more comments.
6	Modular Design	Code is split into reusable functions/modules	Yes	Access-check logic moved into a Script Include and called from ACL scripts.
7	No Redundant Code	Duplicate or unused code is removed	Yes	Duplicate role checks merged; debug gs.log() statements removed.
8	Error Handling	Exceptions and errors are handled gracefully	Partial	try/catch added in main check; access is denied by default if an error occurs.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	Field-value access check (Script Include)	JavaScript (ServiceNow)	All ACL scripts that restrict records by field value	High
2	Role validation helper	JavaScript (ServiceNow)	Read, write and create ACLs	High
3	ACL condition script	JavaScript (ServiceNow)	Other tables with similar field-based restriction	Medium
4	Test cases / test data script	JavaScript (ServiceNow)	Performance and functional testing	Medium

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	4	Consistent indentation and logical organisation of scripts.
Readability	4	Clear variable and function names; logic is easy to follow.
Reusability	4	Core check is in a Script Include and reusable across ACLs.
Documentation / Comments	3	Key logic is commented; some helper functions still need comments.
Overall Score	4 / 5	Well-structured, readable and reusable code with minor scope for better comments.